

Supplementary Material: Additional comments on the paper

Monte Carlo cross-validation in the assessment of Alzheimer's disease diagnoses through machine learning algorithms

The vast majority of studies using machine learning algorithms present the metrics obtained in a single simulation as if they were deterministic values of the developed techniques and use these quantities as a measure of comparison with other algorithms. However, this does not seem to us to be a correct approach (to put it euphemistically) from the point of view of probability theory. Objective metrics are random quantities: each simulation for a set of training and test data generates different results. Even using the same set of test and training data, performance results vary from one simulation to another. Monte Carlo cross-validation is particularly useful because it treats model performance as a **random variable**, rather than as a fixed “pseudo”-deterministic quantity obtained from a single train–test split. In biomedical classification problems, each random partition of the database produces a different training set and a different test set; consequently, the estimated accuracy, sensitivity, specificity, precision, F1-score, false-negative rate, balanced accuracy, and negative predictive value may vary from one simulation to another. Therefore, reporting only one value from a single simulation can lead to an optimistic or pessimistic interpretation of the algorithm’s true behavior. By repeating the experiment many times, Monte Carlo cross-validation allows each metric to be described statistically, through its mean, standard deviation, confidence interval, and probability density function. In this way, the best algorithm is not necessarily the one that produces the highest isolated accuracy, but the one that combines high mean performance with low dispersion, narrow confidence intervals, low false-negative rate, and stable behavior across repeated simulations.

We consider that the use of AI techniques to support medical diagnosis accounts for 7% of the solution to the problem. The other 93% of the solution is associated with the quality of the database. It needs to be a teaching tool that correctly teaches the problem. More importantly, Monte Carlo cross-validation is also useful for evaluating the **database itself**. If the performance metrics show high variability across simulations, this may indicate that the dataset has unstable class boundaries, limited sample representativeness, class imbalance, redundant variables, or strong overlap among the class-conditional distributions. In the context of Alzheimer’s disease diagnosis, this is especially relevant because the separation between cognitively normal individuals and Alzheimer’s disease patients may be easier than the separation between mild cognitive impairment and Alzheimer’s disease, or between cognitively normal individuals and mild cognitive impairment. Thus, Monte Carlo cross-validation does not only rank algorithms; it also reveals the degree to which the available clinical, cognitive, imaging, biomarker, demographic, and genetic variables are sufficient to support reliable diagnostic discrimination. For this reason, the method provides an inference framework for identifying both the most appropriate machine learning techniques and the intrinsic limitations of the database used for classification.

Furthermore, Monte Carlo cross-validation reduces the risk of overfitting by repeatedly estimating the predictive performance of a model under different random partitions of the dataset. Instead of reporting a single deterministic accuracy value, the method provides a distribution of performance metrics, allowing the estimation of both the expected generalization performance and its variability. Models that exhibit large discrepancies between training and test performance, or high variability across simulations, are likely to be overfitted to specific data partitions. Therefore, Monte Carlo cross-validation supports more reliable model selection by favoring algorithms that maintain stable performance across independent resampling experiments.

Although the proposal compares several different algorithms, the objective of the proposal is not limited to this. The following premises guided this research:

- 1 – Evaluate the performance and stability of high-performance algorithms with low dispersion of results (high average accuracy and low standard deviation);
- 2 – Investigate the database's ability to represent the spectral signatures associated with the investigated phenomenon. The low quality of the database in describing the classification object would be evident if the standard deviation were large for all investigated algorithms.
- 3 - Investigate the use of Monte Carlo cross-validation to also investigate/minimize the risk of overfitting.

Translational relevance to biomedical technology

The translational relevance of the proposed framework lies in its ability to move Alzheimer's disease machine learning research beyond a retrospective ranking of algorithms and toward an uncertainty-aware strategy for biomedical decision support. Alzheimer's disease diagnosis remains clinically challenging because the disease evolves progressively from cognitively normal states through mild cognitive impairment to dementia, and the intermediate stages often show overlapping clinical, cognitive, imaging, and biomarker profiles. In this context, the proposed Monte Carlo cross-validation framework provides a clinically meaningful way to evaluate not only the average performance of classifiers, but also the stability, dispersion, confidence intervals, and probability distributions of diagnostic metrics across repeated simulations. Therefore, the framework contributes to biomedical technology by identifying algorithms that are not only accurate, but also robust and reproducible under different train-test partitions, which is essential for the development of reliable clinical decision-support systems.

The intended diagnostic use case is an auxiliary computational tool to support the discrimination of cognitively normal individuals, patients with mild cognitive impairment, and patients with Alzheimer's disease using heterogeneous data derived from clinical practice and biomedical assessment. The study uses variables with direct clinical relevance, including structural MRI volumetric measures, cognitive and functional scales, PET biomarkers, demographic information, and APOE4 genetic status. Thus, the proposed framework could assist neurologists, geriatricians, radiologists, and multidisciplinary memory clinics by supporting patient stratification, early identification of individuals at higher risk, and objective comparison of diagnostic decision boundaries such as CN vs. AD, CN vs. MCI, MCI vs. AD, and CN vs. MCI vs. AD. This is particularly important in the MCI stage, where diagnostic uncertainty is greater and where an evidence-based computational estimate may help guide follow-up, referral, or complementary biomarker testing.

For clinical implementation, the framework should be interpreted as a decision-support component rather than as an autonomous diagnostic system. In a real healthcare environment, it could be integrated into electronic health records, neuroimaging platforms, or memory-clinic workflows, receiving patient-level clinical, cognitive, imaging, biomarker, demographic, and genetic variables as input and returning a diagnostic probability or class-support estimate together with uncertainty measures. Before clinical adoption, however, the framework should be externally validated using independent cohorts, tested prospectively in real-world clinical settings, and evaluated for calibration, interpretability, bias, false-negative behavior, and impact on physician decision-making and patient outcomes. Such validation would determine whether the proposed uncertainty-aware model-selection strategy can improve diagnostic consistency, support earlier detection, and provide a safe pathway for translation into biomedical software for Alzheimer's disease assessment.

This work proposes a validation methodology designed to bridge the gap between machine-learning research and real-world clinical decision support systems. To move from retrospective research to clinical adoption, the following validation stages could be adopted:

- **Phase 1 – External validation:** Test the framework on independent cohorts (e.g., other longitudinal dementia datasets or hospital registries) to assess generalizability.
- **Phase 2 – Prospective observational study:** Deploy the system in memory clinics without influencing clinical decisions and compare:
 - AI predictions
 - Clinician diagnosis
 - Follow-up outcomes
- **Phase 3 – Clinical impact study:** Evaluate whether AI assistance improves:
 - Time to diagnosis
 - Diagnostic agreement
 - Use of advanced biomarkers
 - Patient outcomes and care planning
- **Phase 4 – Regulatory and real-world deployment:** Prepare documentation for regulatory approval as a medical decision-support tool and evaluate performance in routine care.

The Dataset

The final dataset was constructed from the **TADPOLE_D1_D2.csv** file. After selecting the variables of interest, removing records with missing values, retaining only the most recent visit for each participant, and excluding clinically

regressive diagnostic transitions, the final dataset contained **835 unique subjects**, distributed as follows: **CN = 291**, **MCI = 396**, and **AD = 148**. The four diagnostic scenarios were therefore defined as: CN vs. AD, CN vs. MCI, MCI vs. AD, and CN vs. MCI vs. AD. The manuscript already describes the use of ADNI/TADPOLE-derived clinical, cognitive, demographic, MRI, PET, and genetic variables, as well as the one-sample-per-subject strategy used to reduce longitudinal dependence.

We also clarified that the experiments were performed using Monte Carlo cross-validation with **1,000 simulations**, with **80% of the data used for training and 20% for testing** in each simulation. In order to prevent data leakage, all preprocessing operations that depend on the data distribution, including scaling or normalization when required by the algorithm, were performed within each Monte Carlo iteration using only the training partition and then applied to the corresponding test partition. This clarification was added because the manuscript already discusses the need to avoid leakage when preprocessing steps such as scaling, imputation, feature selection, or oversampling are used within resampling-based validation.

Comments on data leakage and circularity

A SHAP (SHapley Additive exPlanations) analysis, a tool used to interpret machine learning models and identify which features contribute most strongly to diagnostic predictions, revealed that cognitive assessments accounted for approximately 75.8% of the model’s decision-making process. In contrast, magnetic resonance imaging-derived biomarkers contributed 14.6%, positron emission tomography (PET) biomarkers contributed 3.9%, and demographic/genetic/educational variables, including APOE4, age, and years of education, contributed 5.7%.

These findings indicate that the model’s predictions were predominantly driven by cognitive and functional measures, whereas neuroimaging and other biological markers played a secondary role. Given that several cognitive variables, such as MMSE, CDRSB, ADAS, FAQ, and RAVLT, are closely related to the clinical criteria used to assign diagnostic labels (CN, MCI, and AD) in ADNI/TADPOLE, the observed feature-importance profile suggests that the model may partially reproduce the underlying clinical diagnostic structure embedded in the DX label, rather than relying exclusively on disease-specific neurobiological signatures.

A sensitivity analysis for the CN vs. AD classification task was performed by partitioning the dataset features into five configurations: the **complete** feature set, the feature set **excluding MRI**-derived variables, the feature set excluding PET-derived variables, the feature set **excluding cognitive variables**, and the feature set excluding **other** variables, namely demographic, genetic, and educational features. In addition, the analysis was conducted across selected classification models, and performance was evaluated using accuracy, balanced accuracy, F1-score, and area under the receiver operating characteristic curve (AUC).

Modelo	Features	Accuracy	Balance Accuracy	F1 Score	AUC
Logistic Regression	complete	99.55	99.21	99.43	1.00
	without MRI	99.55	99.21	99.43	1.00
	without PET	99.77	99.67	99.74	1.00
	without other	99.77	99.67	99.74	1.00
	without cognition	92.50	91.01	91.45	0.96
Random Forest	complete	99.55	99.52	99.54	1.00
	without MRI	99.77	99.76	99.77	1.00
	without PET	99.77	99.62	99.72	1.00
	without other	99.77	99.76	99.77	1.00
	without cognition	91.81	89.85	90.55	0.96

The results of the sensitivity-analysis table demonstrated that cognitive and functional assessments were the main predictors of diagnostic status. The removal of magnetic resonance imaging (MRI), positron emission tomography (PET), or demographic/genetic variables produced negligible changes in performance, whereas the exclusion of cognitive measures resulted in a consistent reduction of approximately 7–8 percentage points in accuracy, balanced accuracy, and F1-score across all classifiers.

However, the models retained high discriminatory performance ($AUC \approx 0.96$) even in the absence of cognitive variables, indicating that neuroimaging biomarkers and clinical risk factors preserve substantial independent diagnostic information. These findings suggest that, although the classifiers partially reproduce the clinical structure used to define the ADNI/TADPOLE diagnostic labels, a significant proportion of their predictive power is derived from biological markers that are not directly involved in the diagnostic-label assignment process.

Several recent studies [1–4] have used feature subsets that are highly similar to the ADNI/TADPOLE feature set employed in the present work, including structural MRI biomarkers, cognitive assessments, PET-derived biomarkers, genetic information, age, and years of education. Cognitive biomarkers are frequently identified as the main contributors to the predictive performance of these models [4].

In the ADNI/TADPOLE dataset used in this study, the assignment of diagnostic labels and their relationship with predictive variables, such as MMSE, CDRSB, ADAS, and FAQ, followed specific criteria designed to ensure clinical consistency. In the present study, diagnostic labels were derived from the original DX variable in the ADNI database, which reflects the participant's clinical status at each visit. The original diagnostic categories were grouped into three final classes: CN, MCI, and AD, including cases of stable MCI and transitions from CN to MCI, as well as cases of stable dementia and transitions from MCI to dementia. To ensure consistency, so-called regressive transitions, such as dementia to MCI or MCI to CN, were removed from the dataset in order to reduce ambiguity and ensure that the labels reflected the most advanced clinical stage observed. Finally, a one-sample-per-participant strategy was adopted, in which the most recent visit of each individual was selected to represent the final clinical status considered in the study.

To prevent data leakage at the individual level, the dataset was constructed using only one visit per participant. The training and test sets were then separated to ensure that the information used to train the model was not used to predict diagnoses in the test set. In addition, the attributes included in this study were selected based on their clinical relevance and on well-established evidence in the literature supporting their use in differentiating among CN, MCI, and AD [5]. This selection strategy aimed to ensure that none of the attributes merely represented a consequence of the outcome to be predicted and that the target outcome was not implicitly or explicitly embedded in any single attribute.

There is an intrinsic overlap between the predictive variables, such as MMSE, CDRSB, ADAS, and FAQ, and the criteria used to assign diagnoses in ADNI clinical practice. Although the manuscript uses these variables as input data, or attributes, to train machine learning algorithms, they are in fact among the same tools used by clinicians in real-world settings to determine whether a patient presents with CN, MCI, or AD [5,6]. Clinical diagnosis is challenging precisely because these cognitive and functional profiles often overlap, particularly in intermediate stages of the disease. The aim of this study is to use artificial intelligence to identify subtle patterns within these variables that may help automate and support this differentiation, while also accounting for the uncertainty caused by such overlap.

Variables such as MMSE and CDRSB exhibit very low overlap values between CN and AD, namely 0.129 and 0.021, respectively, making them label-defining variables in this scenario. Conversely, the results show that MCI classification is the most challenging task due to the high intersection observed for these same scales; for example, MMSE presents an intersection value of 0.932 between CN and MCI.

Additional references

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